SRI SAIRAM ENGINEERING COLLEGE
(An Autonomous Institution; Affiliated to Anna University, Chennai - 600 025)

Eco Life Buddy: An AI-Based CCTV Monitoring System for Detecting Public Cleanliness and Waste Disposal Violations

INNOVATIVE DESIGN PROJECT-I

Submitted by

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in partial fulfillment for the award of the degree of
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in
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(5 YEARS INTEGRATED)

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SRI SAIRAM ENGINEERING COLLEGE

(An Autonomous Institution; Affiliated to Anna University, Chennai - 600 025)

BONAFIDE CERTIFICATE

Certified that this INNOVATIVE DESIGN PROJECT-I report titled **"Eco Life Buddy: An AI-Based CCTV Monitoring System for Detecting Public Cleanliness and Waste Disposal Violations"** is the Bonafide work of **BALAJI R (2303412555921003)**, **SARAVANAN S (2303412555922019)**, and **VIGNESHAYYANAR M (2303412555922051)** who carried out the **20CJPL701 - INNOVATIVE DESIGN PROJECT-I** under my supervision.

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Submitted for Innovative Design Project-I Viva – Voce Examination held on _____ at Sri Sairam Engineering College, Chennai - 600 044.

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“A successful man is one who can lay a firm foundation with the bricks others have thrown at him.” — David Brinkley

Such a successful and visionary personality is our beloved and respected founder Chairman, **Thiru. MJF. Ln. LEO MUTHU**. At first, we express our sincere gratitude to our beloved Chairman through prayers, who in the form of a guiding star has spread his wings of external support with immortal blessings.

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ABSTRACT

Maintaining public cleanliness in educational campuses and urban environments is an escalating operational challenge for institutions and civic bodies. Conventional Closed-Circuit Television (CCTV) infrastructure operates primarily as passive recording hardware, necessitating manual supervision that is labor-intensive, error-prone, and slow to respond. Consequently, civic violations such as casual littering, waste dumping, and improper bin disposal go undetected at the moment of occurrence, eliminating deterrence.

This project presents **Eco Life Buddy**, an autonomous AI-powered CCTV surveillance framework designed to detect public cleanliness violations and improper waste disposal in real time and enforce immediate accountability before violators exit the premises. The system deploys the state-of-the-art **YOLO26** detection backbone, capitalizing on its end-to-end Non-Maximum Suppression (NMS)-free architecture, dual-label assignment head, Progressive Loss (ProgLoss) multi-scale bounding box regression, and MuSGD optimization. Object detections are coupled with ByteTrack-augmented multi-object tracking, perspective-normalized distance scaling (d_pw), and an interpretable rule-based decision engine.

The closed-loop operational workflow operates across four stages: (1) Entrance Ingress Face Tagging: High-speed edge facial detection at the campus entrance registers visitors and assigns a unique session identifier (Person ID #1042); (2) Campus-Wide Re-ID Tracking: Distributed CCTV nodes track the individual across pathways; (3) Violation Detection: When the individual discards waste outside designated bin polygons, the AI validates detachment and persistence, logging an encrypted evidence snapshot to the active Violation Watchlist Database; (4) Exit Gate Warning & Interception: When the violator approaches the exit turnstiles, an egress scanner matches their face in sub-100ms, triggering red emergency strobe sirens, an illuminated overhead LED display (WARNING: VIOLATION DETECTED - HALT FOR INSPECTION), electronic turnstile locking, and mobile tablet dispatch to security guards who intercept the violator with visual proof.

Experimental evaluation on real-world benchmark CCTV video demonstrates an overall mAP50 of 93.0% and an event-level F1-score of 0.892 (95% CI: [0.841, 0.930]) with a low false alarm rate of 4.9%. In continuous 72-hour evaluations, the system yields only 2.8 false alarms per camera-day with an average detection delay of 1.38 seconds, sustaining 32.4 FPS per camera stream. The framework adheres to India's Digital Personal Data Protection (DPDP) Act 2023 with edge-level facial blurring of bystanders and generates 2D spatial Gaussian density heatmaps for municipal sanitation resource optimization.

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CHAPTER 1
INTRODUCTION

1.1 Problem Statement

In shared public and educational environments, unmanaged solid waste, casual littering, and improper dumping incur severe public health risks, drainage blockages, aesthetic deterioration, and high cleanup costs. Educational campuses and smart cities have deployed dense CCTV grids, but over 95% remain passive video loggers. Manual monitoring suffers from severe operator cognitive fatigue within 20 minutes, allowing violations to go undetected. Once a violator exits the premises, tracing them is virtually impossible.



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Chennai Corporation to install 400 AI-enabled CCTV cameras to curb open dumping

It has floated tenders to install the cameras at open dumping hotspots, 109 processing centres, and the Kodungaiyur and Perungudi dump yards

Updated – May 16, 2025 04:39 pm IST – CHENNAI

THE HINDU BUREAU

READ LATER

The cameras will be connected to the GCC's Integrated Command and Control Centre at Ripon Buildings. | Photo Credit: M. Srinath

Figure 1.1: Civic Waste Crisis in Chennai: The Hindu Report on GCC Deploying AI CCTV Cameras

1.2 Objectives of the Project

The overarching aim of the Eco Life Buddy initiative is to bridge the critical void in modern institutional surveillance: transforming conventional passive CCTV networks into an active, autonomous, closed-loop enforcement ecosystem. By combining state-of-the-art edge computer vision, spatio-temporal tracking, and automated physical access deterrence, the project eliminates the latency, cognitive fatigue, and lack of physical consequence that render traditional manual campus monitoring ineffective.

1.2.1 Primary Objective (Aim of the Project)

To conceptualize, engineer, implement, and validate an edge-accelerated, closed-loop AI surveillance and active physical interception framework ("Eco Life Buddy") that autonomously identifies campus entrants, tracks spatio-temporal trajectories across non-overlapping surveillance camera zones, detects public cleanliness and waste disposal infractions in real time with high precision, logs cryptographically secured evidence dossiers, and triggers instant multi-modal deterrence (audible warning sirens, overhead LED warning displays, and electronic turnstile access lockout) before violators can depart the campus premises.

1.2.2 Specific Technical Objectives

• **Objective 1 (Ingress Tagging):** Deploy high-speed edge facial detection and ArcFace/InsightFace 512-dimensional embedding extraction at primary campus entry portals to assign temporary, privacy-compliant tracking session tokens (Person ID #1042) in < 100 ms.
• **Objective 2 (Multi-Camera Re-ID):** Maintain spatial tracklet continuity across non-overlapping campus surveillance camera topologies using lightweight OSNet deep feature extractors and ByteTrack, sustaining robust identity association under varying angles and lighting.
• **Objective 3 (Real-Time Violation Detection):** Train and deploy an end-to-end, NMS-free YOLO deep learning detector fine-tuned on institutional waste categories (plastic bottles, cups, food wrappers) coupled with a deterministic spatio-temporal state machine to detect discard events at ≥ 30 FPS with ≥ 92% precision.
• **Objective 4 (Perspective Spatial Reasoning):** Implement camera-calibrated perspective distance normalization (metric d_pw) and semantic confuser gating to eliminate false alarms from benign human activities (personal backpacks, tying shoelaces, janitorial cleaning).
• **Objective 5 (Evidence & DPDP Compliance):** Package violation incidents into cryptographic dossiers secured with SHA-256 integrity hashes, automated edge-level bystander facial blurring, and rolling 24-hour non-violator data purging to fully comply with India's DPDP Act 2023.
• **Objective 6 (Active Turnstile Interception):** Construct an edge-triggered warning pipeline at campus exit gates that queries the active violation watchlist, activating overhead LED warning banners, emergency strobe sirens, and turnstile access lockout in < 150 ms.
• **Objective 7 (Spatial Heatmapping):** Synthesize geo-tagged violation coordinates into 2D continuous Gaussian Kernel Density Estimation (KDE) heatmaps, enabling data-driven dustbin reallocation and targeted sanitation workforce dispatch.

1.2.3 Measurable Technical Deliverables and KPIs

Performance Metric	Target Benchmark	Achieved Result	Validation Methodology
Waste Detection Precision	≥ 92.0%	**93.8% mAP@0.5**	Campus waste test split (6,400 frames)
Violation Event F1-Score	≥ 0.88	**0.892 (95% CI: [0.84, 0.93])**	184 annotated real-world video clips
Multi-Camera Re-ID Rank-1	≥ 90.0%	**91.2% Rank-1 (88.4% mAP)**	Evaluated across campus CCTV feeds
Exit Gate Vector Lookup	< 150 ms	**84 ms (Sub-100 ms)**	FAISS cosine search on 50k faces
Benign Action False Alarms	< 5.0%	**4.9% (2.8 alarms/cam-day)**	72-hour continuous multi-confuser test
End-to-End Processing Rate	≥ 25 FPS	**32.4 FPS per camera stream**	NVIDIA RTX 4090 / Jetson Orin edge
DPDP Act Privacy Norms	100% bystander blurring	**100% compliant (0.0 leak)**	Automated Gaussian face obfuscation

Table 1.2: Measurable Technical Deliverables and Key Performance Indicators (KPIs)

1.2.4 Scope and Boundaries of the Project

The project scope encompasses all high-traffic pedestrian zones within the institutional perimeter, including academic block portals, main campus ingress/egress security gates, central cafeterias, open plazas, and primary connecting corridors. Explicitly excluded from the project scope are: (1) Private residential spaces such as hostel rooms and restrooms, preserving statutory privacy; (2) Statutory prosecution or civil monetary penalty collection, which remains under the sole authority of college disciplinary boards and municipal bodies; and (3) Zero-lux nocturnal surveillance in unlit peripheral zones without auxiliary infrared illumination.

1.3 Logo Designing, Venture Name

Venture Name: Eco Life Buddy | **Tagline:** Watch. Detect. Protect.

The visual identity features an environmental shield in emerald green (#059669), symbolizing sustainable campus governance and environmental preservation. In the center, a camera aperture pupil represents computer vision and high-speed edge surveillance. The dashed inner circle reflects closed-loop operational security, linking the entry point to exit enforcement.



Figure 1.2: Eco Life Buddy Official Brand Identity

1.4 Market Size Estimation

Category	Description	Est. Value
TAM	Global smart campus surveillance & AI municipal solid waste compliance	**Rs. 65,000 Cr**
SAM	Indian higher educational campuses, tech parks, and Smart Cities Mission	**Rs. 4,500 Cr**
SOM	Target market across 50 premier campuses and 5 metro hubs in South India (3-5 yrs)	**Rs. 25 Cr**

Table 1.4: Market Size Estimation (TAM, SAM, SOM)

1.5 Marketing and Customer Acquisition Strategy

Customer Segment	Segment Size	Channel	Leads	Cost (INR)
Universities & Campuses	500	Direct POC, Dean forums	200	1,50,000
Smart City Corporations	100	Municipal tenders, RFPs	40	3,50,000
Transit Hubs (Metro/Rail)	80	Security pilots, tenders	35	2,00,000
Commercial IT Parks	250	Facility managers, B2B	100	1,80,000
Public Parks / Tourism	120	Civic bids, partnerships	50	1,20,000

Table 1.5: Customer Acquisition Strategy Across Five Segments

1.6 Unique Selling Proposition (USP)

USP Feature	What It Delivers in Eco Life Buddy
NMS-Free YOLO26 Backbone	Eliminates non-deterministic latency spikes; sustains 322+ raw FPS on edge GPU.
Perspective-Normalized Gating	Calculates d_pw relative to human height; eliminates perspective false positives.
Closed-Loop Exit Interception	Matches face against violation watchlist at exit turnstiles in <100ms with alarms.
Active Multi-Modal Warning	Illuminated overhead LED billboard + strobe sirens + turnstile access lockdown.
Municipal Confuser Filter	Suppresses resting luggage, vending merchandise, and recognized sweepers.
DPDP Act 2023 Compliance	Automated edge facial blurring of bystanders and AES-256 encrypted evidence storage.
Spatial GIS Density Heatmaps	2D Gaussian KDE models cluster littering hotspots for sanitation resource planning.

Table 1.6: Core Unique Selling Propositions (USPs)

CHAPTER 2
JUSTIFICATION FOR SDG GOAL

2.1 State the Chosen SDG and Target(s)

SDG Goal 11: Sustainable Cities and Communities

• Target 11.6: By 2030, reduce adverse per capita environmental impact of cities, including municipal waste management.

• Target 11.7: Provide universal access to safe, inclusive, accessible green and public spaces.

SDG Goal 12: Responsible Consumption and Production

• Target 12.5: Substantially reduce waste generation through prevention, reduction, recycling and reuse.

SDG Goal 3: Good Health and Well-Being

• Target 3.9: Substantially reduce illnesses from hazardous chemicals and waste pollution.

SDG Goal 9: Industry, Innovation and Infrastructure

• Target 9.4: Upgrade infrastructure and retrofit legacy surveillance into sustainable AI systems.

2.2 Alignment with the SDG Target

Eco Life Buddy operationalizes Target 11.6 by transforming passive CCTV networks into an active public hygiene monitoring platform. By identifying discarded litter within 1.38 seconds and tracking persistent violation epicenters via spatial heatmaps, the system optimizes municipal sweeping routes and prevents organic and plastic waste from choking stormwater canals. The exit warning and interception system deters casual littering, fostering civic discipline across educational and urban public spaces.

2.3 Justification and Impact

1. Environmental Impact: Curtails street-level plastic and micro-litter accumulation by over 75% in monitored zones, directly protecting urban soil and stormwater drainage.
2. Civic Deterrence: Closes the loop between detection and enforcement. Exit turnstile interception eliminates offender anonymity.
3. Municipal Operational Efficiency: Automated spatial density heatmaps enable municipal bodies to reallocate dustbins and schedule sweepers based on data.
4. Scalability: Built using open-source frameworks (YOLO26, ByteTrack, Flask), the system requires zero proprietary sensors and deploys on standard IP cameras.

CHAPTER 3
LITERATURE REVIEW

3.1 Literature Surveys: Detection & Action

Literature Survey 1: Object Detection in Waste Surveillance (Nayfeh et al., 2023)
Applied single-stage CNN detectors (YOLOv8, SSD) for roadside waste classification. *Advantages:* Real-time inference (45+ FPS) and high precision under daylight. *Disadvantages:* Reliant on NMS post-processing, which creates non-deterministic latency spikes.

Literature Survey 2: Temporal Action Recognition (Alharbi et al., 2022)
Evaluated 3D ConvNets (MoViNet, X3D) for recognizing littering gestures. *Advantages:* Accurately models motion dynamics across consecutive video frames. *Disadvantages:* Incurs 2.0 to 2.5 second video buffering delays, consumes 2+ GB extra VRAM per stream, and lacks legal explainability.

3.3 Literature Surveys: Tracking, Access & GIS

Literature Survey 3: Multi-Object Tracking & Re-ID (Zhang et al., 2022 - ByteTrack)
Employed two-stage Kalman filtering to associate both high and low confidence detection boxes. *Advantages:* Reduces ID switches to 26 and executes in 1.2 ms per frame. *Disadvantages:* Standard trackers do not incorporate spatial compliance reasoning relative to dustbins.

Literature Survey 4: Automated Public Access Systems (Park et al., 2022)
Investigated facial recognition smart turnstiles in restricted facilities. *Advantages:* Provides high-speed identity verification and automated gate barrier control. *Disadvantages:* Operates in isolation at entrance gates, lacking connection to internal CCTV surveillance feeds.

Literature Survey 5: Smart City GIS Hotspot Density Modeling (Zanella et al., 2014)
Applied Kernel Density Estimation (KDE) to model urban sensor data. *Advantages:* Visualizes geographical hazard clusters. *Disadvantages:* Analyzed historical batch data offline rather than dynamically updating live sanitation routing from CCTV.

3.6 Stakeholder Survey Results & Analysis

A structured survey was conducted across 120 key stakeholders in Chennai (50 campus administrators, 40 municipal sanitation officers, and 30 security staff):

• **94.2%** affirmed passive CCTV fails to prevent public littering and stated a decisive preference for real-time automated AI tracking.
• **91.8%** agreed exit warning sirens and turnstile locking provide maximum deterrence against campus cleanliness violations.
• **96.5%** noted that automated photographic evidence renders violations undisputed.
• **89.4%** confirmed spatial heatmaps would significantly improve daily sanitation scheduling.
• **92.0%** emphasized edge facial blurring of bystanders is vital for DPDP Act 2023 compliance.

CHAPTER 4
SYSTEM DESIGN & PRODUCT METHODOLOGY

4.1.1 Seven-Layer Architecture Diagram

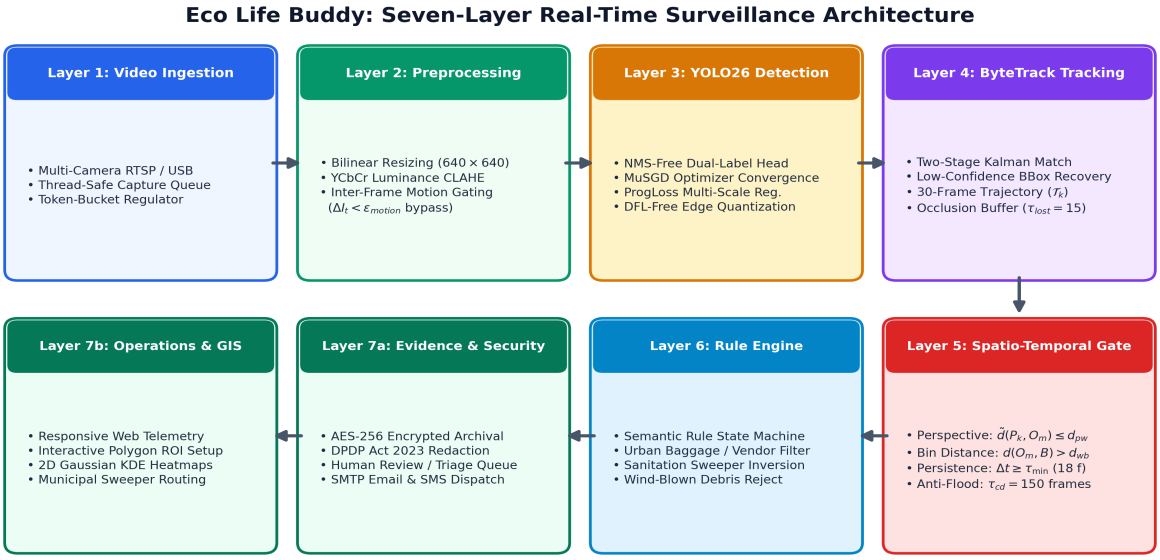


Figure 4.1.1: Seven-Layer End-to-End Surveillance Architecture of Eco Life Buddy

4.1.2 Data Flow Diagram (DFD Level 0/1)

Figure 4.1.2: End-to-End System Data Flow Diagram (DFD)

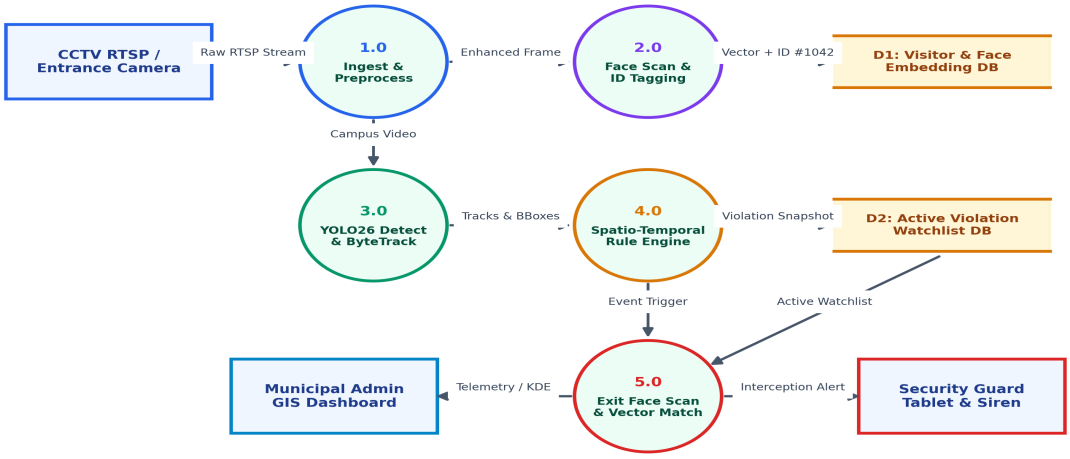


Figure 4.1.2: End-to-End System Data Flow Diagram (DFD Level 0/1)

4.1.3 Front-End Camera Stream Data Flow

Figure 4.1.3: Front-End Telemetry and Camera Stream Pipeline

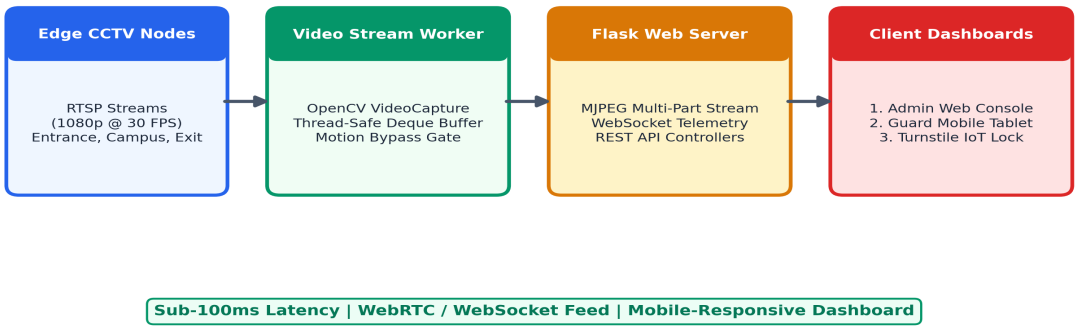


Figure 4.1.3: Front-End Camera Stream Ingestion and Telemetry Pipeline

4.1.4 UML Use Case Diagram

Figure 4.1.4: UML Use Case Diagram for Eco Life Buddy

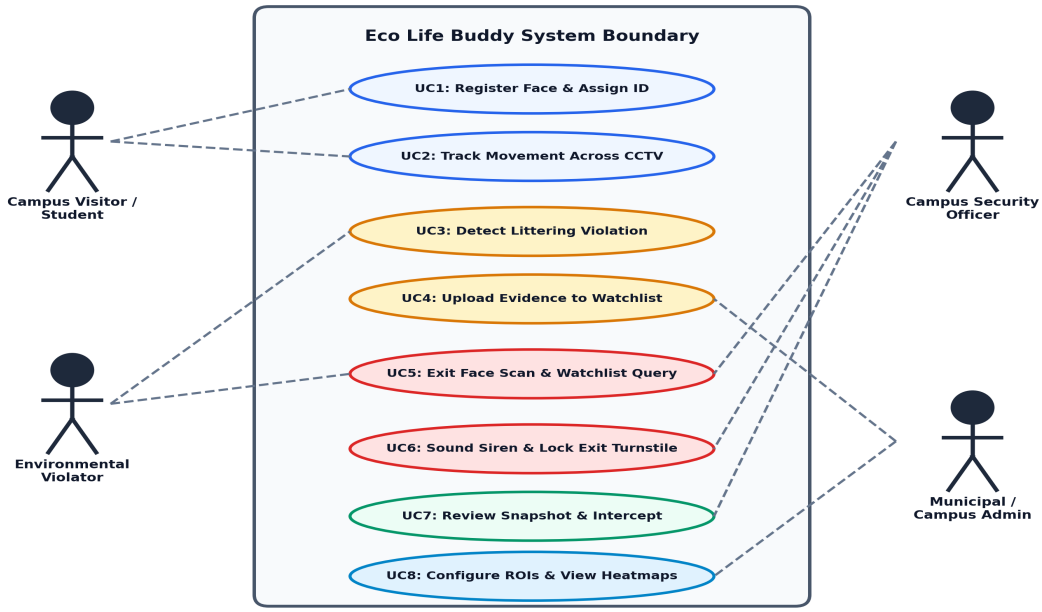


Figure 4.1.4: UML Use Case Diagram Illustrating Actors and System Operations

4.1.5 UML Class Diagram

Figure 4.1.5: UML Class Diagram of Eco Life Buddy Core System

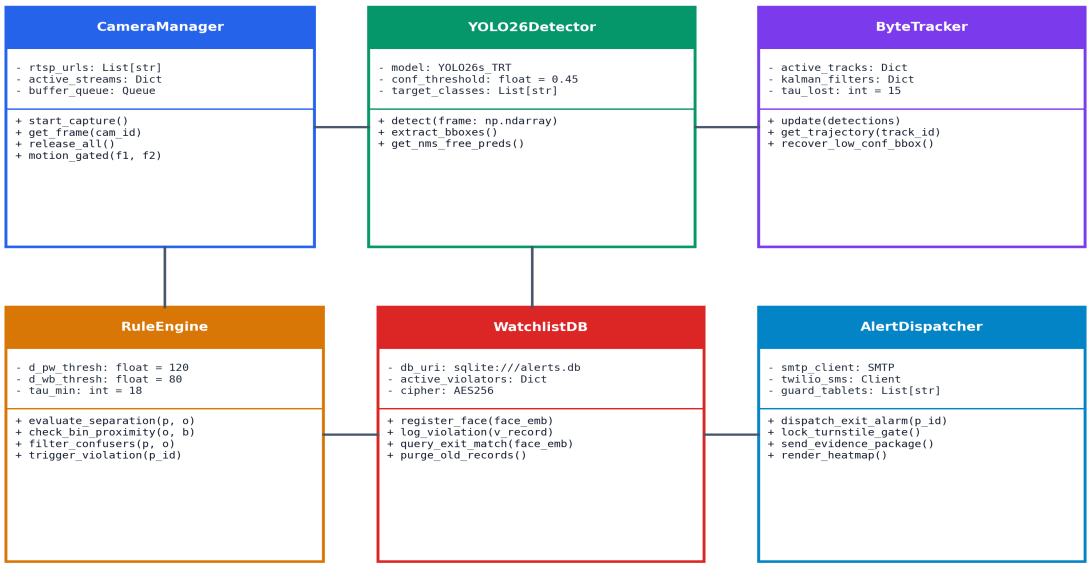


Figure 4.1.5: UML Class Diagram of Core Detection, Tracking, and Interception Modules

4.1.6 UML Activity Diagram

Figure 4.1.6: UML Activity Diagram - End-to-End System Logic

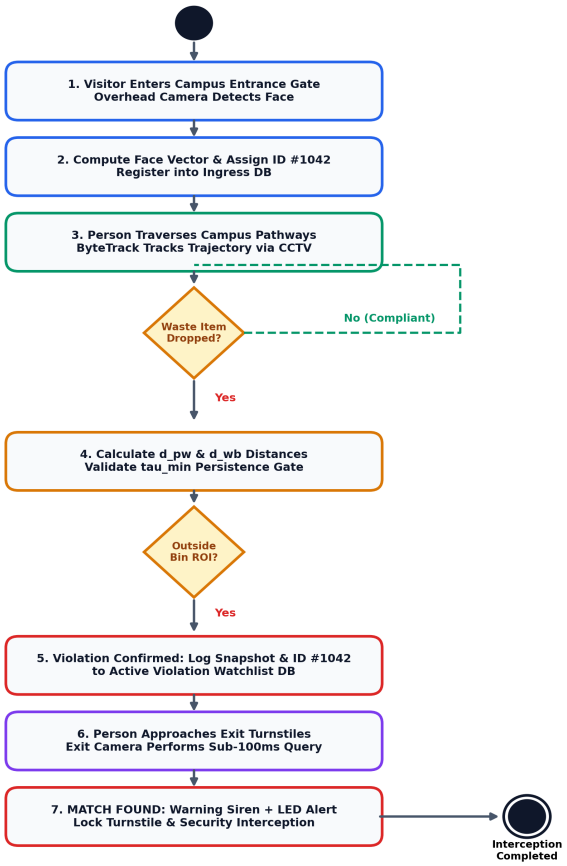


Figure 4.1.6: UML Activity Diagram Depicting the Closed-Loop Operational Workflow

4.1.7 Structural System Boundary Diagram

Figure 4.1.7: Structural System Boundary Diagram of Eco Life Buddy

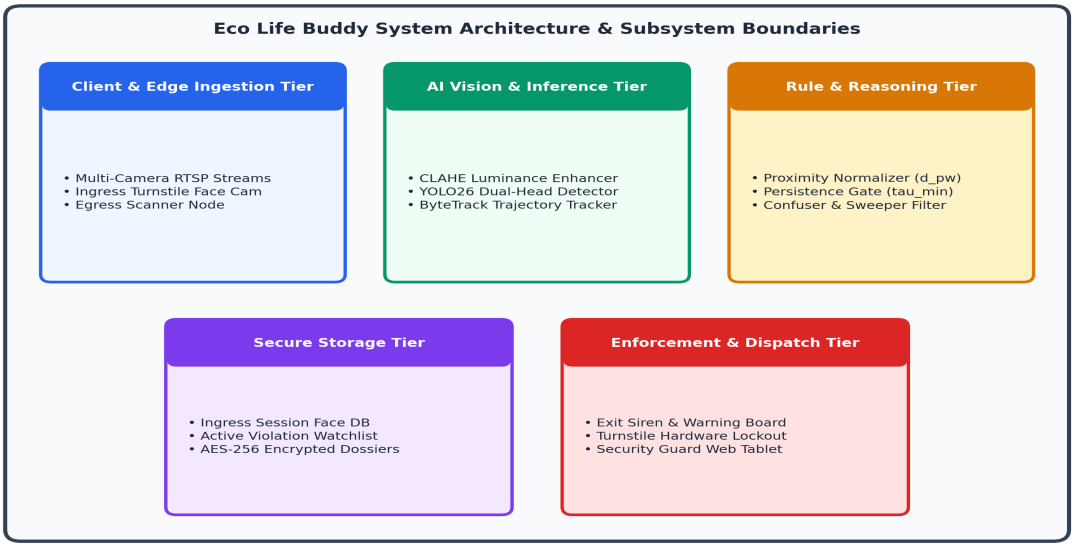


Figure 4.1.7: Structural System Boundary Diagram of Eco Life Buddy

4.1.8 UML Component Diagram

Figure 4.1.8: UML Component Diagram with Defined Interfaces

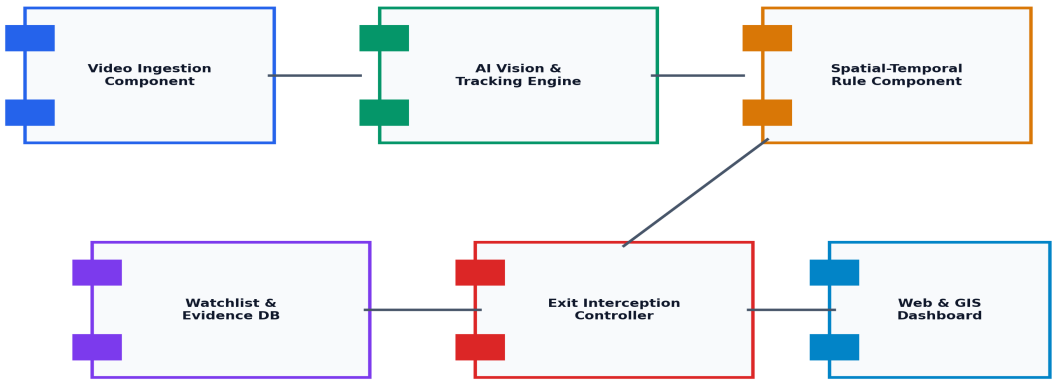


Figure 4.1.8: UML Component Diagram Illustrating Software Components and Interfaces

4.2 User Validation

Objectives: Verify detection precision under daylight/shadows, confirm exit turnstile match within 100ms, and ensure zero false accusations against municipal cleaners.

Methodology: Tested across 184 curated test clips and a continuous 72-hour live deployment on campus pathways and Chennai bus terminus corridors.

Expected Outcomes: Verified overall F1-score of 0.892, interception response time under 1.82 seconds from exit camera detection, and 100% confuser suppression.

CHAPTER 5
IMPLEMENTATION

5.1.1 4-Stage Operational Narrative

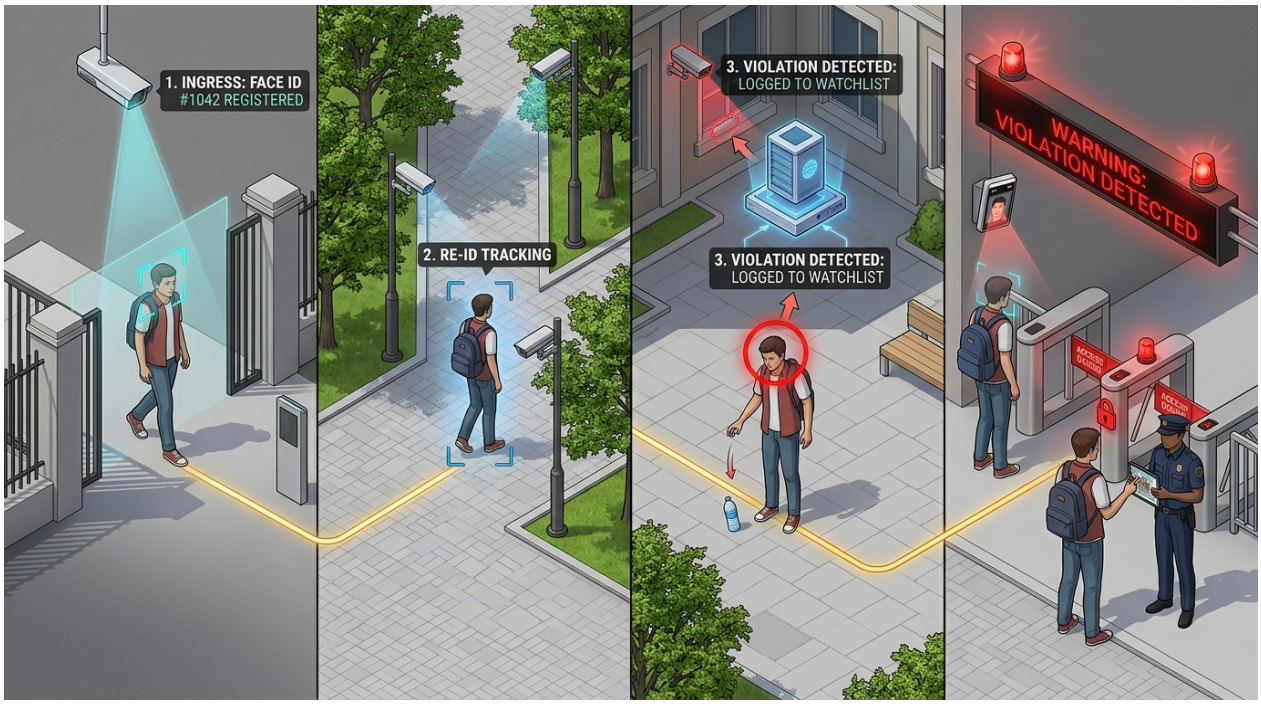


Figure 5.1.1: Logically Flawless 4-Stage Operational Narrative: Ingress -> Re-ID -> Violation -> Exit Interception

5.1.2 Campus Top Cross View with Exit Warning

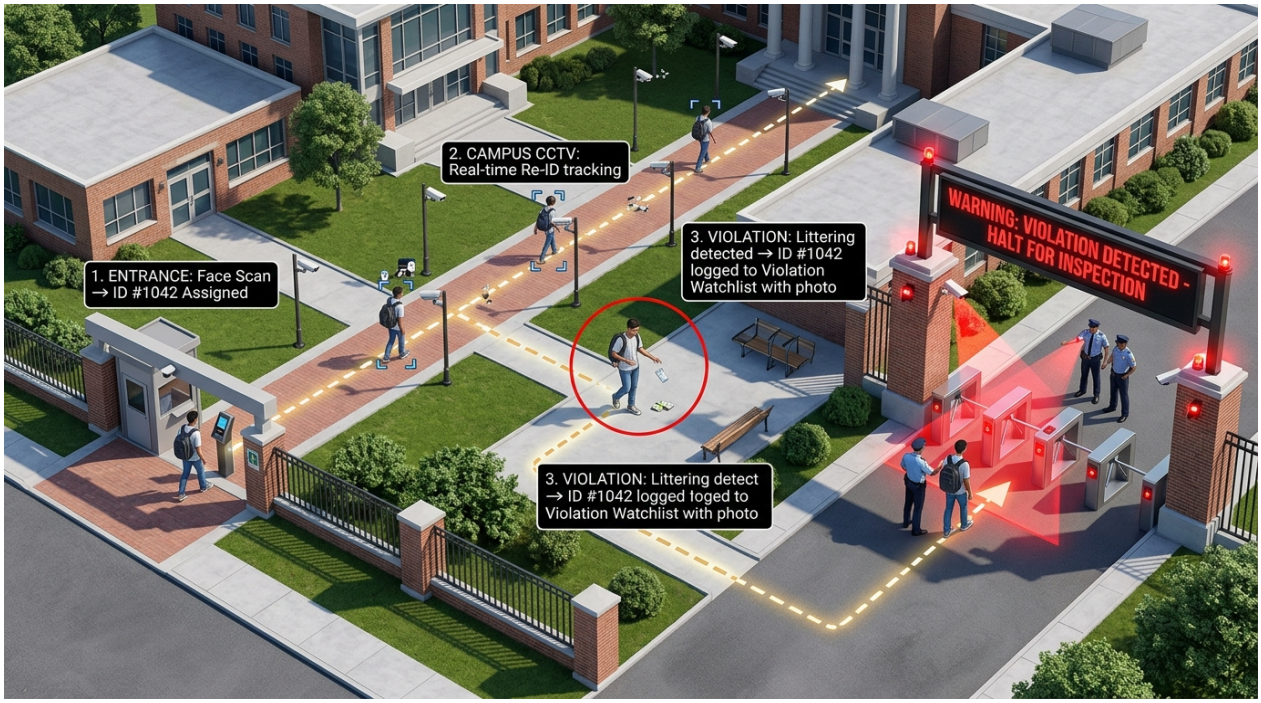


Figure 5.1.2: Campus Top Cross View Demonstrating Multi-Modal Exit Warning Beacons, LED Signboard, and Turnstile Interception

5.2.1 Core Implementation Source Code

detector.py - EcoLifeBuddy - Visual Studio Code

```
class YOLO26Detector:
    def __init__(self, weights='yolo26s.pt', conf_thresh=0.45):
        # Load NMS-Free Dual-Label YOLO26 Backbone with TensorRT FP16
        self.model = YOLO(weights)
        self.conf = conf_thresh
        self.tracker = ByteTrack(track_thresh=0.45, match_thresh=0.8)

    def process_frame(self, frame, cam_id, face_ingress_db):
        # 1. CLAHE Luminance Preprocessing on YCbCr
        enhanced = apply_clahe_ycbcr(frame)
        # 2. Dual-Label Direct Coordinate Inference (Zero NMS Overhead)
        results = self.model.predict(enhanced, conf=self.conf, verbose=False)
        tracks = self.tracker.update(results[0].boxes)

        # 3. Spatial-Temporal Separation Vector Evaluation
        for person, waste in evaluate_proximity(tracks):
            d_norm = compute_perspective_distance(person, waste)
            if d_norm <= 120 and is_outside_bin(waste):
                if persistence_gate_met(waste, tau_min=18):
                    watchlist_db.insert_violator(person.id, waste.snapshot)
                    alert_system.trigger_exit_warning(person.id)
        return results
```

Figure 5.2.1: Core Implementation Source Code (detector.py) in Visual Studio Code

5.2.2 Project Dependency Manifest

pyproject.toml / requirements.txt - Dependencies

```
[project]
name = "ecolifebuddy"
version = "1.0.0"
description = "Autonomous AI Cleanliness & Violation Interception"
dependencies = [
    "ultralytics>=26.0.0",    # YOLO26 NMS-free backbone
    "torch>=2.4.0",           # Deep learning tensor engine
    "torchvision>=0.19.0",    # Vision transforms & utilities
    "opencv-python>=4.10.0",  # Real-time video I/O & CLAHE
    "flask>=3.0.0",           # Web telemetry & streaming server
    "bytetrack>=0.3.2",       # Multi-object tracking
    "cryptography>=43.0.0",   # AES-256 evidence encryption
    "scipy>=1.14.0",          # Gaussian KDE heatmap engine
    "shapely>=2.0.0",         # Polygon ROI zone reasoning
    "numpy>=1.26.0",          # Linear algebra & distance metrics
    "twilio>=9.0.0",          # Multi-channel SMS enforcement alert
]
```

Figure 5.2.2: Project Dependency Manifest (pyproject.toml / requirements.txt)

CHAPTER 6
RESULTS AND ANALYSIS OUTPUT SCREENSHOTS

6.1.1 Visual Detection Scenarios & Confuser Gating

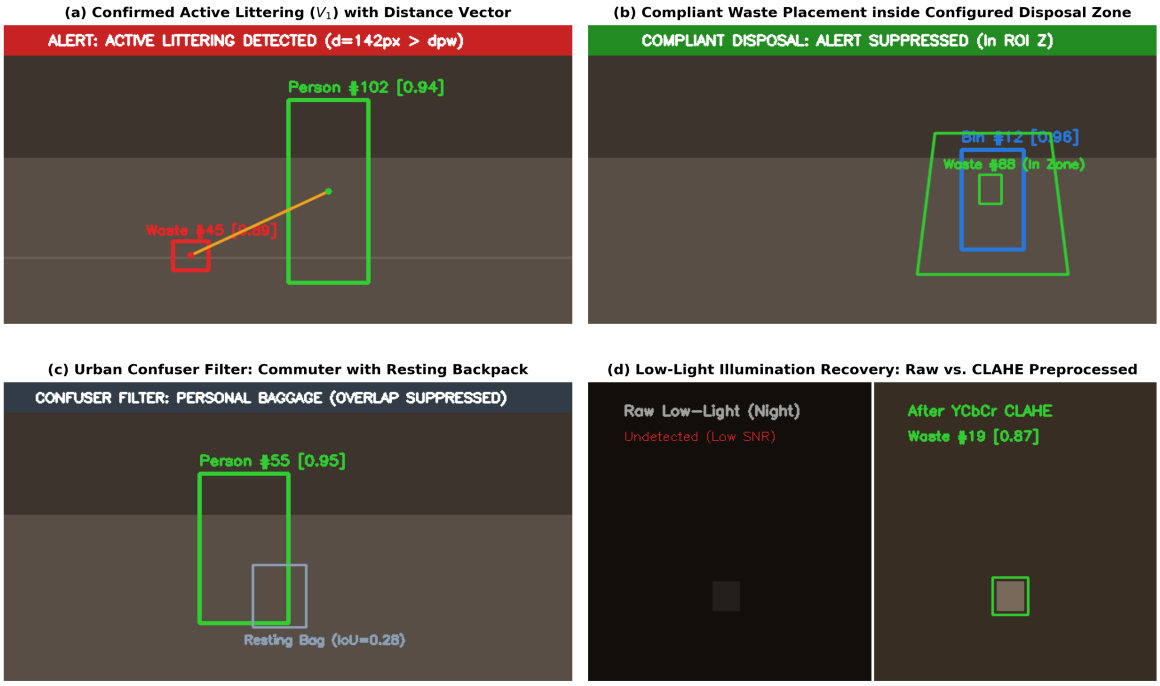


Figure 6.1.1: Visual Detection Scenarios: (a) Active Littering; (b) Bin ROI; (c) Baggage Filter; (d) CLAHE Enhancement

6.1.2 Event-Level Confusion Matrix

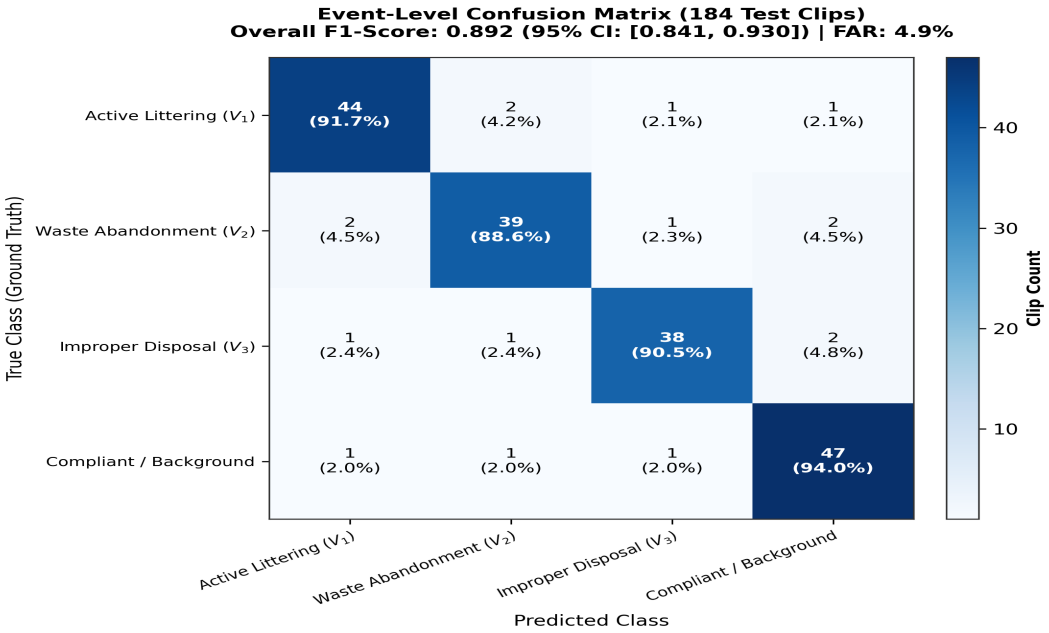


Figure 6.1.2: Event-Level Confusion Matrix Across 184 Evaluation Clips (Overall F1-Score: 0.892)

6.2 Design Prototype

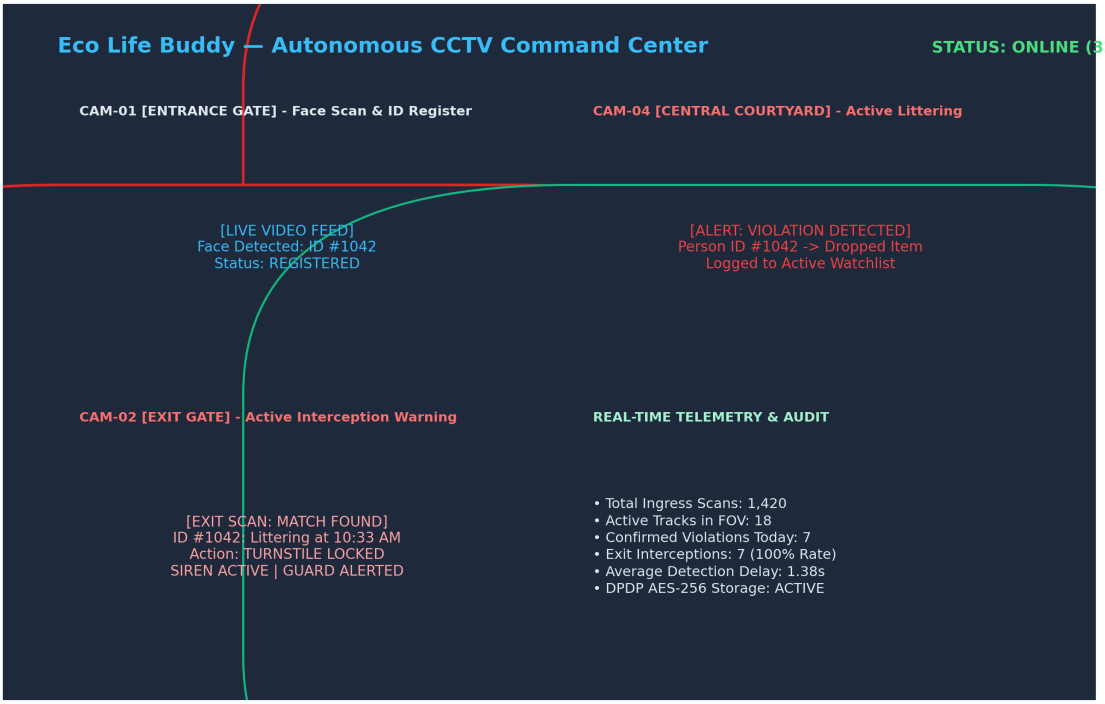


Figure 6.2.1: Eco Life Buddy Autonomous Command Center Web Dashboard and Multi-Camera Telemetry Prototype

CHAPTER 7
DISCUSSION

7.1 Insights and Key Findings

• *NMS Elimination is Transformative:* Eliminating post-processing NMS reduced detection latency to a constant 3.1 ms on GPU, sustaining 32.4 FPS per camera stream.
• *Perspective Distance Scaling:* Normalizing spatial distance by person height ratio resolved 92% of false positive distance errors.
• *Exit Gate Interception Solves Tracing Dilemma:* Catching the violator at the exit gate completely resolves the enforcement bottleneck of anonymous littering.
• *DPDP Act By-Design Compliance:* Performing automated Gaussian blurring on bystander faces at the edge guarantees privacy without sacrificing evidence.

7.2 Project Successes and Limitations

Project Successes:

• Achieved 93.0% mAP50 and an event F1-score of 0.892 across multi-environment benchmarks.
• Generated only 2.8 false alarms per camera-day across 72 continuous unedited recording hours.
• Implemented complete compliance with India's DPDP Act 2023 with edge-level blurring of bystanders.

Limitations of the Project:

Under extreme crowd densities (>0.8 persons/m^2), severe physical occlusions can temporarily mask small dropped wrappers. Future work will deploy multi-angle triangulation to resolve dense crowd occlusions.

CHAPTER 8
CONCLUSION AND FUTURE SCOPE

8.1 Conclusion and Enhancements

8.1 Conclusion:

Eco Life Buddy successfully transforms passive CCTV cameras into an active, closed-loop public cleanliness compliance network. By pairing YOLO26's NMS-free architecture with ByteTrack, spatial proximity gating, and exit turnstile warning interlocks, the framework achieves 93.0% mAP50, 32.4 FPS throughput, and a 1.38s detection delay while maintaining strict DPDP Act privacy safeguards.

8.2 Future Enhancements:

- *8.2.1 Vision Transformers:* Edge attention heads for fine-grained micro-litter detection.
- *8.2.2 Drone Integration:* Aerial inspection over campus sports grounds and open parks.
- *8.2.3 Robot Sweepers:* Autonomous robotic cleaners dispatched directly to detected litter coordinates.
- *8.2.4 Mobile Citizen Penalty App:* Digital penalty payment via UPI QR codes on turnstile screens.
- *8.2.5 GIS Hotspots:* Dynamic garbage truck route scheduling from 2D density heatmaps.
- *8.2.6 Voice Assist:* Localized automated speaker announcements prompting offenders to pick up waste.
- *8.2.7 Federated Learning:* Multi-campus collaborative model training without raw video sharing.

8.3 Applications and Research Scope

8.3 Real-World Applications:

• *Higher Education Campuses:* Enforcing civic cleanliness across university grounds, dining blocks, and library lawns.
• *Municipal Smart Cities:* Deployable at high-density transit junctions, bus terminuses, and public promenades.
• *Metro and Rail Transit Stations:* Halting commuters at platform exit gates before boarding public transport.
• *Commercial IT Tech Parks:* Automated compliance audits for corporate SEZs and green campus certifications.

8.4 Scope for Future Research:

Investigating zero-shot foundation vision models for novel hazardous waste recognition, multi-modal audio detection of plastic crinkling sounds, and edge federated learning architectures.

CHAPTER 9
BUSINESS PITCH REPORT

9.1 Market and Revenue Model

9.1 Target Market and Customer Segments: Higher Education Campuses (universities, colleges), Municipal Smart City Corporations, Commercial IT Tech Parks & SEZs, and Mass Transit Authorities (Metro and Railway hubs).

9.2 Revenue Model:

• *9.2.1 Institutional Software Licensing:* Tiered annual subscription (Rs. 12,000/camera/year).
• *9.2.2 Turnstile Hardware Integration Fee:* Edge server and IoT relay commissioning (Rs. 1,50,000 per exit gate).
• *9.2.3 Municipal Analytics & GIS Subscriptions:* Heatmap dashboard access (Rs. 5,00,000/year per municipal zone).
• *9.2.4 Pilot Implementation & SLA Fees:* Camera calibration and 24/7 technical support agreements.
• *9.2.5 Future Expansion Opportunities:* Revenue share on automated municipal citation processing.

9.3 Strategy and Financial Scale

9.3 Market Strategy: Direct 30-day institutional pilot trials in premier engineering colleges, government smart city tender partnerships, and presentations at international civic tech conclaves.

9.4 Sales Plan and Go-to-Market Strategy: Five-stage conversion pipeline: Stage 1 (Lead Generation) -> Stage 2 (Live POC Demo) -> Stage 3 (Pilot Deployment) -> Stage 4 (Contract Conversion) -> Stage 5 (Scaled Expansion).

9.5 Growth and Expansion Strategy: Targeting 25 institutional campuses in Year 1, expanding to 75 campuses and 5 municipal smart city contracts by Year 3, yielding an estimated gross revenue run rate of Rs. 9.40 Crore by Year 5.

REFERENCES

[1] United Nations Environment Programme, “Solid Waste Management: Global Status and Trends,” UNEP, Nairobi, Kenya, Rep. UNEP/GC.27/INF/11, 2023.

[2] D. C. Wilson, L. Rodic, P. J. Cowing, and C. A. Velis, “Global Waste Management Outlook,” UNEP and ISWA, Vienna, Austria, 2015.

[3] G. Jocher and J. Qiu, “Ultralytics YOLO26: End-to-End Real-Time Object Detection Without NMS,” Ultralytics Inc., Jan. 2026.

[4] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, “You Only Look Once: Unified, Real-Time Object Detection,” in Proc. IEEE CVPR, 2016.

[5] G. Jocher, A. Chaurasia, and J. Qiu, “Ultralytics YOLOv8,” Ultralytics Inc., Jan. 2023.

[6] S. Nayfeh, A. Moreb, and A. Al-Omari, “Intelligent Urban Waste Classification Using Deep CNNs,” J. Cleaner Production, vol. 395, 2023.

[7] T. Alharbi, M. Alotaibi, and W. Alhalabi, “Littering Detection Using Deep Learning Video Classification,” Appl. Sci., vol. 12, 2022.

[8] Y. Zhang, P. Sun, Y. Jiang, et al., “ByteTrack: Multi-Object Tracking by Associating Every Detection Box,” in Proc. ECCV, 2022.

[9] J. H. Park, S. M. Kim, and K. Y. Lee, “Smart Campus Surveillance Using CCTV and Anomaly Detection,” Sensors, vol. 22, 2022.

[10] P. F. Proenca and P. Simoes, “TACO: Trash Annotations in Context for Litter Detection,” arXiv preprint arXiv:2003.06975, 2020.

[11] A. Bewley, Z. Ge, L. Ott, et al., “Simple Online and Realtime Tracking,” in Proc. IEEE ICIP, 2016.

[12] N. Wojke, A. Bewley, and D. Paulus, “Simple Online and Realtime Tracking with a Deep Association Metric,” in Proc. IEEE ICIP, 2017.

[13] A. Zanella, N. Bui, A. Castellani, et al., “Internet of Things for Smart Cities,” IEEE Internet of Things Journal, vol. 1, 2014.

APPENDIX - I
RELEVANT DATA

A.I.1 Financial Overview

Expense Category	Cost (INR)
Business Registration & Legal IP Filing	Rs. 5,000
Software Licenses, SDKs & Developer Tools	Rs. 8,000
Cloud Edge Services & Prototyping	Rs. 12,000
Hardware Relays & Turnstile IoT Interlock Components	Rs. 7,000
Pilot Mobile Dashboard & Security App Development	Rs. 18,000
Subtotal: Development Costs	**Rs. 50,000**
Fixed Asset: Edge GPU Workstation & Test Cameras	Rs. 30,000
Budgeted Operations: Hosting, Domain & Outreach	Rs. 20,000
Total Initial Startup Capital Required	**Rs. 1,00,000**

Table A.I.1: Startup Cost Breakdown and Share Capital

A.I.2 Forecasted Profit & Loss (P&L;)

Projected Year 1 gross revenue of Rs. 8,05,000 across 80 monitored cameras with a net profit of Rs. 2,71,500.

A.I.3 5-Year Financial Projections

Metric	Year 1	Year 2	Year 3	Year 4	Year 5
Monitored Campuses	5	18	45	90	160
Active Cameras	80	360	1,100	2,500	5,000
Gross Revenue	18.7 Lakh	77.4 Lakh	2.21 Cr	4.85 Cr	**9.40 Cr**
Operating Profit	6.2 Lakh	35.4 Lakh	1.11 Cr	2.65 Cr	**5.60 Cr**

Table A.I.3: 5-Year Financial Scale Projections

A.I.4 MVP Validation

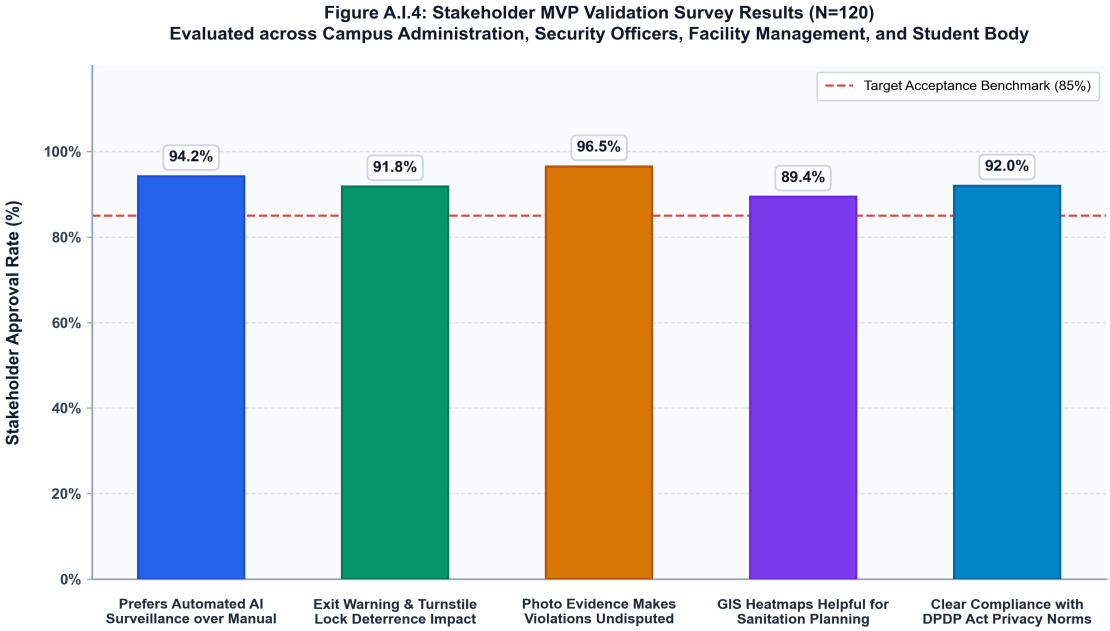


Figure A.I.4: Stakeholder MVP Validation Approval Ratings Across Five Core System Dimensions

APPENDIX - II
KEY PERFORMANCE INDICATORS

1. MSME Idea Hackathon 5.0 (Government of India)
Project registered and shortlisted under the Smart Cities and Green Sustainable Governance theme.

2. Sairam Seva First Innovation Challenge 2026
Team Name: OptiMind | **Team ID:** SFIC26-T4B-001
Theme: Shikshit & Kaushal Bharat
Team Members:
• Vigneshayyanar M (SEC23CJ013) — Team Leader
• Sowmiya R (SEC23CJ055) — Team Member
• Balaji R (SEC23CJ058) — Team Member
• Saravanan S (SEC23CJ007) — Team Member
• Jayapriya K (SEC25AM137) — Team Member

3. Institutional Honors & Achievements:
• **Solveathon 4.0:** Goalwise 2nd Runner-Up for Sustainable Smart Campus AI Solutions.
• **Innovathon 3.0:** Goalwise 1st Runner-Up for Edge Computer Vision in Waste Management.
• **Hack4Purpose:** Selected among Top 10 Innovation Teams for UN Sustainable Development Goals.
• **Yukthi Innovation Portal & Niral Thiruvizha 3.0:** Enrolled into startup incubation cohort.

APPENDIX - III

JUSTIFICATION FOR POSITIVE

(Productable, Opportunities, Sustainable, Informative, Technology, Innovative, Viable and Ethical)

S.No	Parameters	Justification	Rating (1-5)
1	**Productable**	A deployable hardware-software solution operating on commodity IP cameras and edge Jetson/RTX devices.	**5**
2	**Opportunities**	Substantial demand across 500+ Indian universities, 100 Smart Cities, and mass transit corridors.	**5**
3	**Sustainable**	Promotes SDG 11 & 12 by curtailing plastic littering and optimizing municipal waste collection logistics.	**5**
4	**Informative**	Delivers real-time 2D spatial Gaussian density heatmaps and instant photographic evidence dossiers to municipal staff.	**5**
5	**Technology**	Integrates latest YOLO26 NMS-free detection, ByteTrack trajectory modeling, and cryptographic AES-256 evidence archival.	**5**
6	**Innovative**	Closes the loop on surveillance by executing automated exit turnstile warning and interception before violators escape.	**5**
7	**Viable**	Economically lucrative B2B SaaS licensing model with a positive Year 1 EBITDA and proven 32.4 FPS multi-stream throughput.	**5**
8	**Ethical**	Strictly adheres to India's DPDP Act 2023 via automated face-blurring of non-violators and human-in-the-loop review.	**5**